Introduction to (Statistical) Machine Learning
Problem Set Four

Summer 2026, IIM B

Due by 11:59 pm on July 17, 2026.

Instructions. All four problems are required. For theoretical problems, using LLMs is not allowed. You are free to use them for the coding problems.

1. Suppose $X_1, \ldots, X_n$ are independent and identically distributed observations from a distribution with finite variance σ^2.

    a) Describe an algorithm based on the classical bootstrap to obtain a 95% interval for σ.

    b) Describe an algorithm based on the Bayesian bootstrap to obtain a 95% interval for σ.

    c) Suppose now that

    $$X_1, \ldots, X_n \overset{\text{i.i.d}}{\sim} N(\mu, \sigma^2).$$

    Describe a 95% uncertainty interval, frequentist or Bayesian, for σ. Clearly state any prior used if you choose a Bayesian approach.

    d) The file `old_faithful_eruptions.txt` contains eruption durations for the Old Faithful geyser in Yellowstone National Park. Using each of the three methods described above, compute a 95% interval for σ.

    Compare the resulting intervals. Are the intervals similar? Which method produces the widest interval? Briefly explain the differences you observe.

2. Consider the text data file `alice.txt`. The goal of this problem is to fit two probabilistic language models to the text: an independent and identically distributed model and a first-order Markov, or bigram, model.

    a) Tokenize the file `alice.txt` so that every token corresponds to a single word, with the period, or full stop, treated as a separate token.

    b) Construct a vocabulary consisting of all unique tokens in the corpus. Let K denote the size of the vocabulary.

    c) In the first model, assume that the tokens

    $$Y_1, \ldots, Y_n$$

    are independent and identically distributed from a categorical distribution with probabilities

    $$(p_1, \ldots, p_K).$$

Use the prior

$$(p_1, \dots, p_K) \sim \text{Dirichlet}(\alpha, \dots, \alpha),$$

where α is small, for example $\alpha = 10^{-3}$.

Calculate the posterior distribution of $(p_1, \dots, p_K)$.

i. Report the five tokens with the largest posterior mean probabilities. Comment on whether the result is reasonable given the nature of the text.

ii. Using this model, generate 20 new sentences. To generate each sentence, independently generate tokens until the period token is produced, and treat the period as the end of the sentence.

   Display the generated sentences and comment on how realistic they appear.

d) We next use a first-order Markov, or bigram, model. Let $x_{j|i}$ denote the number of times token j immediately follows token i in the corpus. The likelihood is

$$\prod_{i=1}^{K} \prod_{j=1}^{K} p_{j|i}^{x_{j|i}},$$

where

$$p_{j|i} = \mathbb{P}\{Y_{t+1} = j \mid Y_t = i\}.$$

For each token i, use the prior

$$(p_{1|i}, \dots, p_{K|i}) \sim \text{Dirichlet}(a_1, \dots, a_K),$$

independently over i.

Using the method of maximum marginal likelihood discussed in class, estimate the hyperparameters

$$a_1, \dots, a_K.$$

e) Using the estimated hyperparameters, calculate the posterior mean estimates of $p_{j|i}$ for every i and j.

For each of the tokens

`alice`, `she`, `the`,

report the four possible succeeding tokens j having the largest estimated values of $p_{j|i}$.

f) Use the fitted bigram model to generate 20 new sentences. Begin each sentence with the period token. Then generate tokens sequentially from the estimated transition probabilities until another period token is generated.

Display the generated sentences. Comment on how realistic they appear and compare them with the sentences generated under the independent model.

3. Consider the monthly Google Trends dataset `GolfTrends13March2026.csv`. The dataset records the search popularity in the United States of the query *golf*, from January 2004 through February 2026.

Let y_t denote the observed value in month t. We consider the model

$$y_t = \mu_t + \epsilon_t, \qquad \epsilon_t \overset{\text{i.i.d}}{\sim} N(0, \sigma^2),$$

where the trend function is represented as

$$\mu_t = \beta_0 + \beta_1(t-1) + \beta_2(t-2)_+ + \beta_3(t-3)_+ + \cdots + \beta_{n-1}(t-(n-1))_+.$$

Here

$$u_+ = \max\{u, 0\}.$$

a) Consider the ridge regression estimator obtained by minimizing

$$\sum_{t=1}^{n} [y_t - \beta_0 - \beta_1(t-1) - \beta_2(t-2)_+ - \cdots - \beta_{n-1}(t-(n-1))_+]^2$$
$$+ \lambda \sum_{j=2}^{n-1} \beta_j^2.$$

Implement this estimator. Use a suitable cross-validation procedure to choose the regularization parameter λ.

Report the selected value of λ and plot the estimated trend $\hat{\mu}_t$ together with the data. Comment on whether the fitted trend captures the important patterns in the data without interpolating all of the observations.

b) Consider Bayesian inference under the prior

$$\beta_0, \beta_1 \overset{\text{i.i.d}}{\sim} N(0, C),$$

and

$$\beta_2, \ldots, \beta_{n-1} \overset{\text{i.i.d}}{\sim} N(0, \gamma^2\sigma^2),$$

where C is very large.

Use a uniform prior on a sufficiently wide bounded interval for $\log \sigma$ and a uniform prior

$$\log \gamma \sim \text{Uniform(low, high)}$$

for suitably chosen values of low and high.

i. Generate 1000 posterior samples of γ and σ. Using these samples, calculate point estimates $\hat{\gamma}$ and $\hat{\sigma}$.

   Compare the value of λ selected by cross-validation in part (a) with

   $$\frac{1}{\hat{\gamma}^2}.$$

   Explain why these two quantities are naturally related.

ii. Generate 1000 posterior samples of the trend

   $$(\mu_1, \ldots, \mu_n).$$

   Plot the sampled trend functions together with the observed data. Comment on whether the uncertainty displayed by the sampled functions appears reasonable.

iii. Average the posterior samples to obtain the Bayesian posterior mean estimate of μ_t. Compare this estimate with the ridge regression estimate from part (a).

Which estimate is smoother? Which estimate would you prefer, and why?

4. Consider again the Google Trends dataset `GolfTrends13March2026.csv`. In the previous problem, the fitted function represented only a smooth long-term trend. The data also exhibit a strong seasonal pattern. We therefore consider the model

$$\begin{aligned} y_t &= \beta_0 + \beta_1(t-1) + \beta_2(t-2)_+ + \cdots + \beta_{n-1}(t-(n-1))_+ \\ &\quad + \beta_n \cos\left(\frac{2\pi t}{12}\right) + \beta_{n+1} \sin\left(\frac{2\pi t}{12}\right) + \epsilon_t, \end{aligned}$$

where

$$\epsilon_t \overset{\text{i.i.d}}{\sim} N(0, \sigma^2).$$

Define the nonseasonal trend component by

$$\mu_t = \beta_0 + \beta_1(t-1) + \beta_2(t-2)_+ + \cdots + \beta_{n-1}(t-(n-1))_+.$$

a) Consider the ridge regression estimator obtained by minimizing

$$\begin{aligned} \sum_{t=1}^{n} \Bigg[& y_t - \beta_0 - \beta_1(t-1) - \beta_2(t-2)_+ - \cdots - \beta_{n-1}(t-(n-1))_+ \\ & - \beta_n \cos\left(\frac{2\pi t}{12}\right) - \beta_{n+1} \sin\left(\frac{2\pi t}{12}\right) \Bigg]^2 + \lambda \sum_{j=2}^{n-1} \beta_j^2. \end{aligned}$$

Notice that the intercept, linear trend, and sinusoidal coefficients are not penalized.

Implement this estimator and use a suitable cross-validation procedure to choose λ. Report the selected value of λ.

On the same plot as the observed data, display both

$$\hat{\mu}_t = \hat{\beta}_0 + \hat{\beta}_1(t-1) + \hat{\beta}_2(t-2)_+ + \cdots + \hat{\beta}_{n-1}(t-(n-1))_+$$

and

$$\hat{\mu}_t + \hat{\beta}_n \cos\left(\frac{2\pi t}{12}\right) + \hat{\beta}_{n+1} \sin\left(\frac{2\pi t}{12}\right).$$

Comment on how well the nonseasonal trend and the complete fitted function capture the patterns in the data.

b) Consider Bayesian inference under the prior

$$\beta_0, \beta_1, \beta_n, \beta_{n+1} \overset{\text{i.i.d}}{\sim} N(0, C),$$

and

$$\beta_2, \ldots, \beta_{n-1} \overset{\text{i.i.d}}{\sim} N(0, \gamma^2\sigma^2),$$

where C is very large.

Use a uniform prior on a sufficiently wide bounded interval for $\log \sigma$ and the prior

$$\log \gamma \sim \text{Uniform}(\text{low}, \text{high})$$

for suitably chosen values of low and high.

i. Generate 1000 posterior samples of γ and σ. Using these samples, calculate point estimates $\hat{\gamma}$ and $\hat{\sigma}$.

   Compare the value of λ selected in part (a) with

$$\frac{1}{\hat{\gamma}^2}.$$

ii. Generate 1000 posterior samples of the nonseasonal trend

$$(\mu_1, \ldots, \mu_n).$$

   Plot these sampled trend functions together with the observed data. Comment on whether the uncertainty displayed in the plot appears reasonable.

iii. Average the posterior samples to obtain the Bayesian posterior mean estimate of μ_t. Compare this estimate with the ridge regression estimate from part (a).

   Which estimate is smoother? Which estimate would you prefer, and why?

iv. Obtain posterior mean estimates of β_n and β_{n+1}. Using these estimates and the posterior mean estimate $\hat{\mu}_t$ from part (iii), plot

$$\hat{\mu}_t + \hat{\beta}_n \cos\left(\frac{2\pi t}{12}\right) + \hat{\beta}_{n+1} \sin\left(\frac{2\pi t}{12}\right)$$

   together with the observed data.

   Comment on how well the fitted function captures both the long-term trend and the seasonal variation in the data.